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Chapter 2: Single Queueing Systems

2.1 Introduction

In this chapter we will look at the case of the single queue. Even though chapters 3 and 4 will discuss networks of queues, it is probably safe to say that the majority of modeling work that has been done for queues has involved a single queue. One reason is that for any investigation into queueing theory the single queueing system is a natural starting point. Another reason is tractability: much more can easily be ascertained about single queues than about networks of queues.

We will start by examining the background of the most basic of queueing systems, the M/M/1 queue. This ''Markovian'' queue is distinguished by a Poisson arrival process and exponential service times. It is also distinguished by a one dimensional, and infinite in extent, state transition diagram. The coordinate of the one dimensional state transition diagram is simply the number of customers in the queueing system.

Later, by modifying the rates associated with this diagram and the diagram's extent we will arrive at a number of other useful queueing systems such as systems with multiple servers and systems with finite waiting line capacity.

We will spend some time looking at a non-Markovian queue of great interest, the M/G/1 queueing system. For this generalized queueing system a surprising amount of analytic information can be obtained.

For some queueing systems it is easier to calculate state probabilities in the z domain rather than directly. We will look at this z-transform technique in terms of two examples, the transient analysis of the M/M/1 queueing system and the M/G/1 queueing system.

Finally, there has been recent interest in systems carrying multiple classes of traffic. In the last section of this chapter the existing literature on multiclass queueing models is reviewed.

2.2 The M/M/1 Queueing System

Consider the queueing system of Figure 2.1. It is the queueing system from which all queueing theory proceeds. This M/M/1 queueing system has two crucial assumptions. One is that the arrival process is a Poisson process, and the other is that the single server has a service time which is an exponentially distributed random variable. These assumptions lead to a very tractable model and are reasonable for a wide variety of situations. We will start by examining these assumptions.

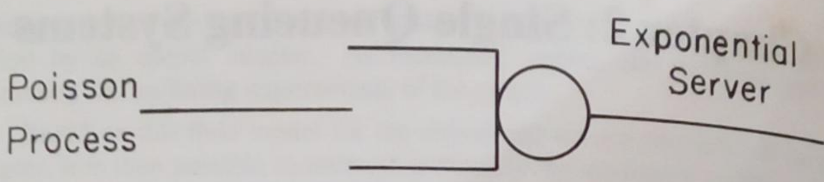

Fig. 2.1: M/M/1 Queueing System

2.2.1 The Poisson Process

The *Poisson process* is the most basic of arrival processes. Interestingly, it is a purely random arrival process. One way of looking at it goes like this. Suppose that the time axis is divided into a large number of small time segments of width Δt. We will let the probability of a single customer arriving in a segment be proportional to the length of the segment, Δt, with a proportionality constant, λ, which represents the mean arrival rate:

$$P(\textit{exactly } 1 \textit{ arrival in } [t,t+\Delta t]) = \lambda\Delta t, \tag{2.1}$$

$$P(\textit{no arrivals in } [t,t+\Delta t]) = 1 - \lambda\Delta t, \tag{2.2}$$

$$P(\textit{more than } 1 \textit{ arrival in } [t,t+\Delta t]) = 0. \tag{2.3}$$

Here we ignore higher order terms in Δt.

Now we can make an analogy between the arrival process and coin flipping. Each segment is like a coin flip with $\lambda\Delta t$ being the probability of an arrival (say, heads) and $1 - \lambda\Delta t$ being the probability of no arrival (say, tails).

As $\Delta t \rightarrow 0$ we form the continuous time Poisson process. From the coin flipping analogy one can see that arrivals are independent of one another as they can be thought of as simply the positive results of a very large number of independent coin flips. Moreover, one can see that no one instant of time is any more or less likely to have an arrival than any other instant.

Two ideas that extend the usefulness of the Poisson process are the concept of a random split of a Poisson process and a joining of Poisson processes. Figure 2.2 illustrates a queueing system with a random split. Arrivals are randomly split between three queues with independent probabilities p_1, p_2, and p_3. A situation involving a joining of independent Poisson processes is illustrated in Figure 2.3. Arrivals from a number of independent Poisson processes are joined or merged to form an aggregate process.

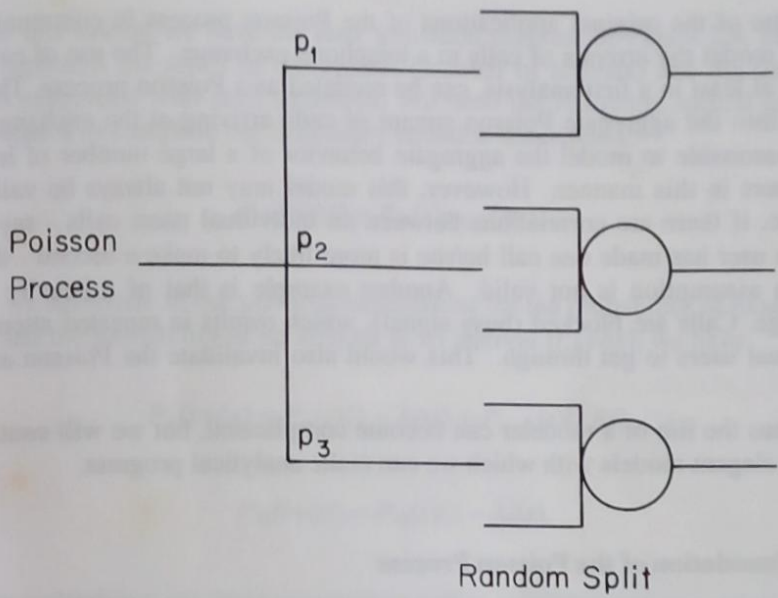

Fig. 2.2: Random Split of a Poisson Process

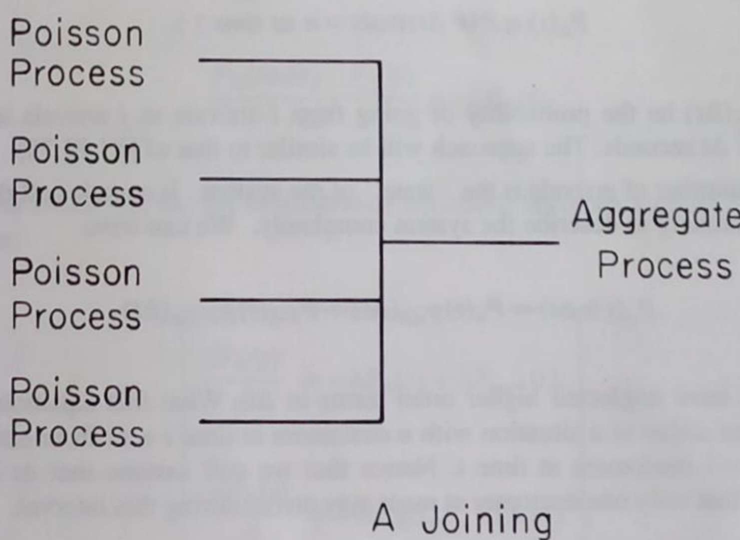

Fig. 2.3: Joining of Poisson Processes

It turns out that random splits of a Poisson process are Poisson and that a joining of independent Poisson processes is also Poisson. A little thought with the coin flipping analogy will show that this is true.

One of the original applications of the Poisson process in communications was to model the arrivals of calls to a telephone exchange. The use of each telephone, at least in a first analysis, can be modeled as a Poisson process. These are joined into the aggregate Poisson stream of calls arriving at the exchange. It is very reasonable to model the aggregate behavior of a large number of independent users in this manner. However, this model may not always be valid. For instance, if there are correlations between an individual users calls - say, given that the user has made one call he/she is more likely to make a second - then the Poisson assumption is not valid. Another example is that of a heavily loaded exchange. Calls are blocked (busy signal), which results in repeated attempts by individual users to get through. This would also invalidate the Poisson assumption.

Thus the life of a modeler can become complicated, but we will continue to discuss elegant models with which we can make analytical progress.

2.2.2 Foundation of the Poisson Process

We will derive the underlying differential equation of the Poisson process by using difference equation arguments and letting $\Delta t \rightarrow 0$. We will start by letting

$$P_n(t) \equiv P(\#\ Arrivals = n \text{ at time } t\ ), \tag{2.4}$$

and let $p_{ij}(\Delta t)$ be the probability of going from i arrivals to j arrivals in a time interval of Δt seconds. The approach will be similar to that of [KLEI 75].

The number of arrivals is the "state" of the system. It contains all the information necessary to describe the system completely. We can write

$$P_n(t+\Delta t) = P_n(t)p_{n,n}(\Delta t) + P_{n-1}(t)p_{n-1,n}(\Delta t). \tag{2.5}$$

Again, we have neglected higher order terms in Δt. What this equation says is that one can arrive at a situation with n customers at time $t+\Delta t$ from either having n or $n-1$ customers at time t. Notice that we still assume that Δt is small enough so that only one customer at most may arrive during this interval.

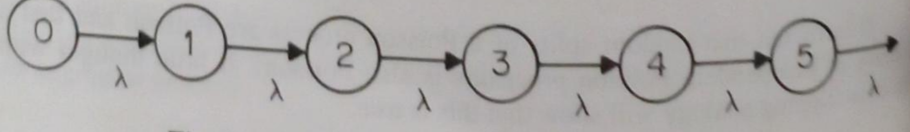

Fig. 2.4: Poisson Process State Transition Diagram

For this system we have the state transition diagram of Figure 2.4. Here the circles represent the states of the system (number of arrivals) and the transition rate λ is associated with each transition. We need the following special equation for the state 0 to complete our difference equation description:

$$P_0(t+\Delta t) = P_0(t)p_{0,0}(\Delta t). \tag{2.6}$$

If we substitute in our previous expressions for the probability of exactly one arrival and the probability of no arrivals in an interval $[t, t+\Delta t]$, we have

$$P_n(t+\Delta t) = P_n(t)(1-\lambda\Delta t) + P_{n-1}(t)(\lambda\Delta t), \tag{2.7}$$

$$P_0(t+\Delta t) = P_0(t)(1-\lambda\Delta t). \tag{2.8}$$

By multiplying out these expressions and re-arranging one can arrive at

$$\frac{P_n(t+\Delta t) - P_n(t)}{\Delta t} = -\lambda P_n(t) + \lambda P_{n-1}(t), \tag{2.9}$$

$$\frac{P_0(t+\Delta t) - P_0(t)}{\Delta t} = -\lambda P_0(t). \tag{2.10}$$

If we let $\Delta t \rightarrow 0$, the set of difference equations becomes a set of differential equations:

$$\boxed{\begin{aligned} \frac{dP_n(t)}{dt} &= -\lambda P_n(t) + \lambda P_{n-1}(t), \\ \frac{dP_0(t)}{dt} &= -\lambda P_0(t). \qquad (2.11) \end{aligned}}$$

Here $n \geq 1$. We now naturally wish to find the solution to these differential equations, $P_n(t)$. For the second equation, a knowledge of differential equations enables us to see that

$$P_0(t) = e^{-\lambda t}. \tag{2.12}$$

This makes the next equation

$$\frac{dP_1(t)}{dt} = -\lambda P_1(t) + \lambda e^{-\lambda t}, \tag{2.13}$$

which has a solution of

$$P_1(t) = \lambda t e^{-\lambda t}. \tag{2.14}$$

Hence the next equation is

$$\frac{dP_2(t)}{dt} = -\lambda P_2(t) + \lambda^2 t e^{-\lambda t}, \tag{2.15}$$

which has a solution of

$$P_2(t) = \frac{\lambda^2 t^2}{2} e^{-\lambda t}. \tag{2.16}$$

Continuing, we would find by induction that

$$\boxed{P_n(t) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}. \qquad (2.17)}$$

This is the *Poisson distribution*. It tells us the probability of n arrivals in an interval of t seconds for a Poisson process of rate λ. A plot of the Poisson distribution appears in Figure 2.5.

Example: A telephone exchange receives 100 calls a minute on average, according to a Poisson process. What is the probability that no calls are received in an interval of five seconds?

The solution is simply

$$P_0(t) = e^{-\lambda t} = e^{-100 \times \frac{1}{12}} = 0.00024.$$

∇

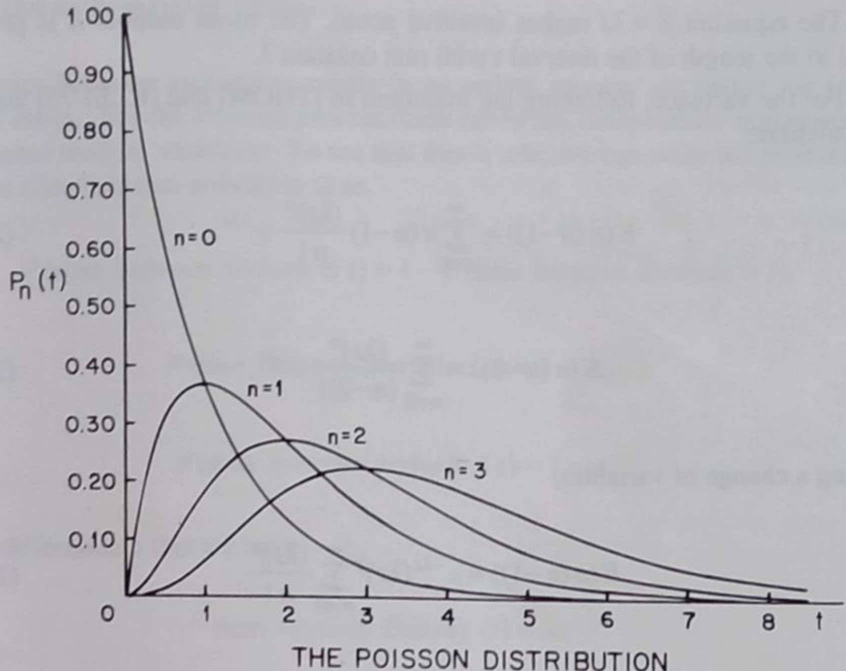

Fig. 2.5: Poisson Distribution

2.2.3 Poisson Distribution Mean and Variance

We will calculate the mean and variance of the Poisson distribution. Let $\bar{n}$ be the mean number of arrivals in an interval of length t. Then:

$$\bar{n} = \sum_{n=0}^{\infty} nP_n(t), \tag{2.18}$$

$$\bar{n} = \sum_{n=0}^{\infty} n \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \tag{2.19}$$

$$\bar{n} = e^{-\lambda t} \sum_{n=1}^{\infty} \frac{(\lambda t)^n}{(n-1)!}. \tag{2.20}$$

With a change of variables this is

$$\bar{n} = e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda t)^{n+1}}{n!}, \tag{2.21}$$

$$\bar{n} = e^{-\lambda t} \lambda t \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} = e^{-\lambda t} \lambda t e^{\lambda t} = \lambda t. \tag{2.22}$$

The equation $\bar{n} = \lambda t$ makes intuitive sense. The mean number $\bar{n}$ is proportional to the length of the interval t with rate constant λ.

For the variance, following the treatment in [THOM] and [KLEI 75] we will first calculate

$$E(n(n-1)) = \sum_{n=0}^{\infty} n(n-1)\frac{(\lambda t)^n}{n!}e^{-\lambda t}, \tag{2.23}$$

$$E(n(n-1)) = \sum_{n=2}^{\infty} \frac{(\lambda t)^n}{(n-2)!}e^{-\lambda t}. \tag{2.24}$$

Making a change of variables,

$$E(n(n-1)) = e^{-\lambda t}(\lambda t)^2 \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!}, \tag{2.25}$$

$$E(n(n-1)) = e^{-\lambda t}(\lambda t)^2 e^{\lambda t} = (\lambda t)^2. \tag{2.26}$$

Now

$$E(n(n-1)) = E(n^2) - E(n) = (\lambda t)^2, \tag{2.27}$$

or

$$E(n^2) = (\lambda t)^2 + \lambda t. \tag{2.28}$$

We also make use of the fact that the variance of n is

$$\sigma_n{}^2 = E(n-\bar{n})^2 = E(n^2) + E(\bar{n})^2 - 2\bar{n}\,E(n), \tag{2.29}$$

$$\sigma_n{}^2 = E(n^2) - \bar{n}^2. \tag{2.30}$$

Substituting one has

$$\sigma_n{}^2 = (\lambda t)^2 + \lambda t - (\lambda t)^2 = \lambda t. \tag{2.31}$$

So the mean and the variance of the Poisson distribution are both equal to λt.

2.2.4 The Inter-Arrival Times

The times between successive events in an arrival process are called the inter-arrival times. For the Poisson process these times are independent exponentially distributed random variables. To see that this is true, we can write the probability that the time between arrivals is $\le t$ as

$$P(\textit{time between arrivals} \le t) = 1 - P(\textit{time between arrivals} > t),$$

$$P(\textit{time between arrivals} \le t) = 1 - P_0(t),$$

$$P(\textit{time between arrivals} \le t) = 1 - e^{-\lambda t}.$$

If we differentiate this we have

$$\textit{Inter-arrival Density}\ (t) = \lambda e^{-\lambda t}.$$

An exponential random variable is the only continuous random variable that has the *memoryless property*. Loosely speaking, this means that the previous history of the random variable is of no help in predicting the future. To be more specific, a complete state description of an M/M/1 queueing system at an instant of time consists only of the number of customers in the system at that instant. We do not have to include in the state description the time since the last arrival because the M/M/1 queueing system has a Poisson arrival process with exponential inter-arrival times. Furthermore, we also do not have to include the time since the last service completion in the state description since the service time in an M/M/1 queueing system is exponentially distributed.

In terms of the Poisson arrival process the memoryless property means that the distribution of time until the next event is the same no matter what point in time we are at. Let us give an intuitive analogy. Suppose that trains arrive at a station according to a Poisson process with a mean time between trains of 20 minutes. Suppose now that you arrive at the station and are told by someone standing at the platform that the last train arrived nineteen minutes ago. What is the expected time until the next train arrives?

One would naturally like to answer one minute, and this would be true if trains arrived deterministically exactly twenty minutes apart. However, we are dealing with a Poisson arrival process. The answer is twenty minutes!

To see that this is true, we have to go back to our coin flipping explanation of the Poisson process. An event is the result of a positive outcome of a "coin flip" in an infinitesimally small interval. As time proceeds there are a very large number of such flips. When we put the Poisson process in this context it is easy to see that the distribution until the next positive outcome does not depend at all on when the last positive outcome occurred. That is, you cannot predict when you will flip heads by knowing the last time when heads occurred.

We can make this idea more precise. If an arrival occurs at time $t=0$, we know from before that the distribution of time until the next arrival is $\lambda e^{-\lambda t}$, $t \geq 0$, and the cumulative distribution function is $1-e^{-\lambda t}$, $t \geq 0$. Suppose now that no arrivals occur up until time t_0. Given this information what is the distribution of time until the next arrival? We can write it in the following way [KLEI 75] with the dummy variable t^*:

$$P(t^* \leq t+t_0 \mid t^* > t_0) = \frac{P(t_0 < t^* \leq t+t_0)}{P(t^* > t_0)}, \tag{2.32}$$

$$P(t^* \leq t+t_0 \mid t^* > t_0) = \frac{\int_{t_0}^{t+t_0} \lambda e^{-\lambda t^*} dt^*}{\int_{t_0}^{\infty} \lambda e^{-\lambda t^*} dt^*}, \tag{2.33}$$

$$P(t^* \leq t+t_0 \mid t^* > t_0) = \frac{-e^{-\lambda t^*} \Big|_{t_0}^{t+t_0}}{-e^{-\lambda t^*} \Big|_{t_0}^{\infty}}, \tag{2.34}$$

$$P(t^* \leq t+t_0 \mid t^* > t_0) = \frac{-e^{-\lambda(t+t_0)} + e^{-\lambda t_0}}{-0 + e^{-\lambda t_0}}, \tag{2.35}$$

$$P(t^* \leq t+t_0 \mid t^* > t_0) = 1 - e^{-\lambda t}. \tag{2.36}$$

This is the same cumulative distribution function we had when we asked what the distribution was at time $t=0$!

We close this section by noting the *discrete* distribution with the memoryless property is the geometric distribution [FELL] (see chapter 6).

2.2.5 The Markov Property

The memoryless property is more accurately referred to as the *Markov property*. It says, intuitively, that one can predict the future state of the system based on the present state as well as if one had the past history of the state available. In terms of a continuous, non-negative, random time T, this Markov property can be expressed as [COOP]

$$P(T>t_0+t \mid T>t_0) = P(T>t) \tag{2.37}$$

for every $t_0>0$ and every $t>0$. In terms of a discrete random variable $X(t_n)$ one can say [KLEI 75] that

$$P(X(t_{n+1})=x_{n+1} \mid X(t_n)=x_n, X(t_{n-1})=x_{n-1}, \ldots\ldots, X(t_1)=x_1)$$

$$= P(X(t_{n+1})=x_{n+1} \mid X(t_n)=x_n) \tag{2.38}$$

where the t_i are monotonically increasing and the x_n take on some values from a discrete state space.

Markovian systems are memoryless systems. This makes them relatively simple to analyze since when one describes the state of the system one does not need to take into account the time since the last arrival or the time that service has been underway. When the arrival process is not Poisson or the service time is not exponential then we must take these quantities into account, This makes understanding the behavior of even a single queue under these conditions a non-trivial undertaking.

A concept that we will come across later is that of a *Markov chain*. This is a Markov process with a discrete state space. The appearance of the state transition diagram of the M/M/1 queueing system which we are about to see (Figure 2.6), accounts for the use of the word ''chain''.

2.2.6 Exponential Service Times

In an M/M/1 queueing system the service times are independent exponentially distributed random variables. That is, they occur according to the probability density $\mu e^{-\mu t}$ where μ is the mean service rate and $1/\mu$ is the mean time for service. As has been discussed, this type of server is memoryless and thus produces a tractable analysis.

How realistic is an exponential server? It has been found to be reasonable for modeling the duration of telephone conversations. But it has also been used in situations where the physical basis for its use is weaker. For instance, the time to transmit fixed length packets of information in computer networks is often modeled by an exponential random variable. Sometimes this sort of assumption is made for reasons of tractability, even if some degree of realism is sacrificed.

The exponential server will be with us for a long time because, in conjunction with the Poisson process, it leads to Markovian systems and the associated elegant network wide results of the next two chapters.

2.2.7 Foundation of the M/M/1 Queueing System

Let us take Figure 2.4 and add transitions that represent servicing to create the state transition diagram of Figure 2.6. This state transition diagram represents what is called a *birth-death process.* That is, if we think of the state as being a population size, it may increase by one member at a time ("birth") or it may decrease by one member at a time ("death"). There are other models that may change by several members at a time (called "batch arrival" or "batch departure" processes). In the context of the performance evaluation of information systems the state of the system can actually be thought of as the number of packets in a communication processor, the number of new calls in a telephone exchange computer, or the number of jobs in a computer. The approach we will take is similar to that in [KLEI 75].

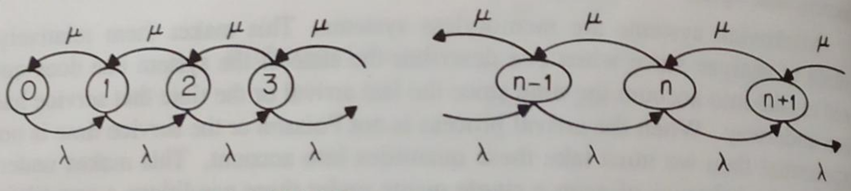

Fig. 2.6: M/M/1 System State Transition Diagram

When we were discussing the Poisson process we had the events

$$P(\textit{exactly 1 arrival in } [t,t+\Delta t]) = \lambda\Delta t,$$

$$P(\textit{no arrivals in } [t,t+\Delta t]) = 1 - \lambda\Delta t.$$

Add to this the following two events:

$$P(\textit{1 service completion in } [t,t+\Delta t]) = \mu\Delta t,$$

$$P(\textit{no service completions in } [t,t+\Delta t]) = 1 - \mu\Delta t.$$

Also again we will define

$$P_n(t) \equiv P(\#\ \textit{Arrivals} = n \textit{ at time } t\,)$$

and let $p_{ij}(\Delta t)$ be the probability of going from i arrivals to j arrivals in a time interval of Δt seconds.

Whereas previously the state of the system was the number of Poisson arrivals, it is now the number in the queueing system, including the one in service. We can write:

$$P_n(t+\Delta t) = \tag{2.39}$$

$$P_n(t)p_{n,n}(\Delta t) + P_{n-1}(t)p_{n-1,n}(\Delta t) + P_{n+1}(t)p_{n+1,n}(\Delta t).$$

Higher-order terms in Δt are neglected.

This equation says that one can arrive at a situation with n customers at time $t+\Delta t$ from either having $n-1$, n or $n+1$ customers at time t. Because this is a birth-death process, these are the only possibilities. For the state 0 we will need the following special equation:

$$P_0(t+\Delta t) = P_0(t)p_{0,0}(\Delta t) + P_1(t)p_{1,0}(\Delta t). \tag{2.40}$$

We will now substitute the previous expressions for the probabilities of the various events in an interval $[t,t+\Delta t]$ into this equation to yield

$$P_n(t+\Delta t) = \tag{2.41}$$

$$P_n(t)(1-\lambda\Delta t)(1-\mu\Delta t) + P_{n-1}(t)(\lambda\Delta t) + P_{n+1}(t)(\mu\Delta t),$$

$$P_0(t+\Delta t) = P_0(t)(1-\lambda\Delta t) + P_1(t)(\mu\Delta t).$$

By multiplying out these expressions, re-arranging and letting $\Delta t \rightarrow 0$ one has

$$\boxed{\begin{aligned} \frac{dP_n(t)}{dt} &= -(\lambda+\mu)P_n(t) + \lambda P_{n-1}(t) + \mu P_{n+1}(t), \\ \frac{dP_0(t)}{dt} &= -\lambda P_0(t) + \mu P_1(t). \qquad (2.42) \end{aligned}}$$

Here $n \geq 1$.

These differential equations describe the evolution in time of the state probabilities of the M/M/1 queueing system. But what do these equations really mean? To answer this question we will need the following discussion of flows and balancing.

2.2.8 Flows and Balancing

One of the most intuitively pleasing concepts in queueing theory is the way in which one can make an analogy between the flow of current in an electric circuit and the flows in the state transition diagram of a queueing system. But what is "flowing" in the state transition diagram?

The answer is that *probability flux* flows from one state to another along the transitions. The probability flux along a transition, numerically, is the product of the probability of the state at which the transition originates and the transition rate associated with the transition. The probability flux along a transition, physically, is the mean number of times per second that the event corresponding to the transition occurs. For instance, if a transition has a rate of 10 sec^{-1} and the probability of the state at which the transition originates is 0.1 then the transition is actually traversed by the system state once a second, on average.

The analogy with electric circuits is straightforward. The state probabilities correspond to nodal voltages, and the transition rates correspond to conductance (inverse resistance) values. There are some differences though. Specifically:

> There is no analog of a voltage source or battery in the queueing world. Rather we have the normalization equation that holds that at any instant of time the sum of the state probabilities sums to one. This normalization equation "energizes" the state transition diagram in the sense that it guarantees that there will be non-zero flows.

> The direction of flow is preset in the state transition diagram by the transition direction. As has been mentioned what "flows" is the product of the probability of the state at which the transition originates and the transition rate. This is different from the electric circuit case where a branch current's magnitude and direction depend on the difference of voltages of the two nodes to which the branch is connected.

The most useful concept that carries over from the electric circuit world to the queueing theory world is the idea of looking at what flows across boundaries in the circuit or state transition diagram. For instance, suppose we draw a boundary around state n in the state transition diagram of the M/M/1 queueing system, as in Figure 2.7. What we are going to do is equate the mean number of times a second the state is entered with the mean number of times one leaves the state. What flows out of the state at time t is

$$-(\lambda + \mu)P_n(t)$$

and what flows into the state from states $n-1$ and $n+1$ is

$$\lambda P_{n-1}(t) + \mu P_{n+1}(t).$$

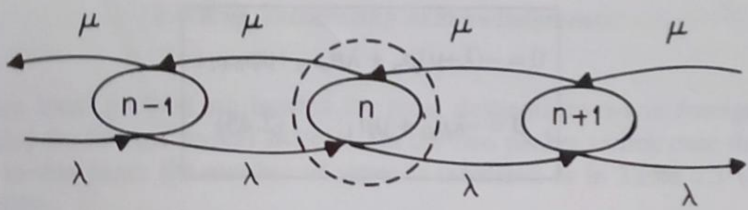

Fig. 2.7: Global Balance Boundary Around a State

The difference between these two quantities is just the change in the probability of state n or:

$$\frac{dP_n(t)}{dt} = -(\lambda+\mu)P_n(t) + \lambda P_{n-1}(t) + \mu P_{n+1}(t). \tag{2.43}$$

But this is just the same differential equation that we found using difference equation arguments in the last section! The equation for $dP_0(t)/dt$ can be derived in a similar manner.

So for the M/M/1 queueing system we have a set of differential equations to describe the evolution of the state probabilities as a function of time. As with electric circuits, this is useful if one wants a *transient analysis* of the queueing system. This is the study of the queueing system over a limited time period. For instance, we may be interested in the queueing system's behavior for the first ten minutes after service commences for an initially empty queue. Then the initial conditions of the set of differential equations becomes

$$P_0(0) = 1.0, \tag{2.44}$$

$$P_n(0) = 0.0, \qquad n \geq 1,$$

and the differential equations can be solved. We will look at this situation in section 2.11.

An interesting thing happens when we let $dP_n(t)/dt = 0$. This is the situation that occurs when the queueing system has been running for a long time, transient behavior has settled out and the queue is in equilibrium. *Equilibrium or steady-state analysis* deals with this situation.

Since the probabilities do not change with time we can write:

$$\boxed{\begin{aligned} 0 &= -(\lambda+\mu)p_n + \lambda p_{n-1} + \mu p_{n+1}, \\ 0 &= -\lambda p_0 + \mu p_1. \qquad (2.45) \end{aligned}}$$

Here $n \geq 1$. Note that we have changed probability to lower case.

These equations are now a set of linear equations rather than a set of differential equations. Each equation can be arrived at by drawing a boundary completely around a specific state and equating what flows into the state with what flows out of the state. An equation arrived at in such a manner is known as a *global balance equation*. The reader familiar with electric circuits will be reminded of Kirchoff's current conservation law which holds that the *net* flow of current into and out of a circuit node equals zero as charge can not build up at a node. Similarly, in equilibrium the mean rate at which a state is entered should equal the mean rate at which the state is left.

We can calculate the equilibrium state probabilities of any queueing system with a finite number of states in the following manner. For each of the N states we write one global balance equation. This gives us a set of N linear equations with N unknowns. Any one of the equations is redundant and should be replaced with the equation

$$p_0 + p_1 + p_2 + \ldots\ldots + p_N = 1,$$

which normalizes the solution. The resultant set of linear equations can then be solved using standard numerical techniques. The reason that we have emphasized the calculation of the state probabilities is that many performance measures of interest are functions of the state probabilities.

The difficulty with this approach is that realistic queueing models often have extremely large state spaces making numerical solution computationally infeasible.

Example: In [KAUF] L. Kaufman, B. Gopinath and E. F. Wunderlich present a Markovian model of packet switches. A packet switch is a dedicated processor which transmits, relays and receives packets of digital information. It requires four dimensions to describe each packet switch in this model:

$$h = \#\ \textit{of local packets in input buffer},$$

$$i = \#\ \textit{of foreign packets in input buffer},$$

$$j = \#\ \textit{of packets in output buffer},$$

$$k = \#\ of\ outstanding\ acknowledgments.$$

Here local packets are headed for local destinations while foreign packets are headed for foreign packet switches. In the two packet switch case that is considered in this paper the number of states is tabulated as in Table 2.1 (copyright 1981 IEEE).

Table 2.1: Number of Model States

Buffer Size	Window Size	6-D Model	8-D Model
4	2	961	4,225
5	2	2,116	12,321
6	2	4,096	30,625
6	3	5,476	38,025
8	4	21,025	211,600
9	4	34,225	416,025

Here the window size refers to the allowable number of outstanding acknowledgments. The growth in the number of states for the two packet switch, eight-dimensional model is impressive given the relatively small buffer and window size. A reduced six-dimensional model is possible if one does not distinguish between the number of local and foreign packets in each switch but only looks at the total number of packets in the input buffer. While this reduces the state space size, modest increases in the buffer and window sizes would negate this savings.

∇

Luckily, there is a large class of useful networks which satisfy a more specific type of balancing that leads to an analytic solution. Suppose we redraw the M/M/1 queueing system state transition diagram as in Figure 2.8.

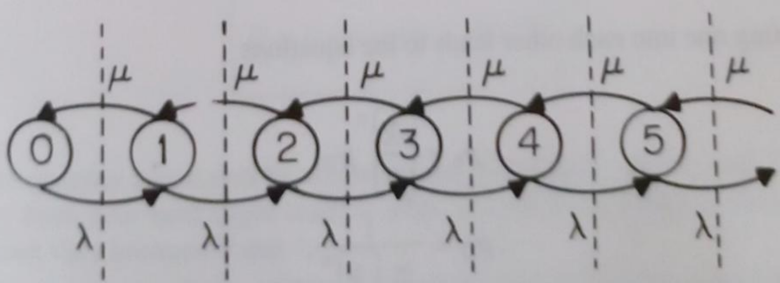

Fig. 2.8: Local Balance Boundaries Between States

Here we have drawn vertical boundaries between each adjacent pair of states. Each boundary actually separates the state transition diagram into two halves. We can equate the flow from left to right across such a boundary with the flow from right to left. We thus obtain a set of *local balance equations*:

$$\lambda p_0 = \mu p_1, \tag{2.46}$$

$$\lambda p_1 = \mu p_2,$$

$$\lambda p_2 = \mu p_3,$$

$$\lambda p_{n-1} = \mu p_n.$$

From these equations we can easily write the recursions:

$$p_1 = \frac{\lambda}{\mu} p_0, \tag{2.47}$$

$$p_2 = \frac{\lambda}{\mu} p_1,$$

$$p_3 = \frac{\lambda}{\mu} p_2,$$

$$p_n = \frac{\lambda}{\mu} p_{n-1}.$$

Substituting one into each other leads to the equations

$$p_n = \left[\frac{\lambda}{\mu}\right]^n p_0, \tag{2.48a}$$

$$p_0 = \frac{1}{\sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^n}. \tag{2.48b}$$

Here we used the normalization (conservation of probability) equation to solve for p_0. This result is the one-dimensional version of what is known as the product form solution for the state probabilities. The generalization appears in chapter 3.

2.2.9 The M/M/1 Queueing System in Detail

We can further simplify the expression for p_0 if we make use of the identity

$$\sum_{n=0}^{\infty} \rho^n = \frac{1}{1-\rho}, \qquad 0 \le \rho < 1, \tag{2.49}$$

so that

$$p_n = \rho^n p_0, \tag{2.50}$$

$$p_0 = (1-\rho), \tag{2.51}$$

or

$$\boxed{p_n = \rho^n (1-\rho) \qquad (2.52)}$$

where we are letting $\rho = \lambda/\mu$. The variable ρ is known as *utilization*. This makes sense since for an M/M/1 queueing system the utilization of the queueing system is the probability that the queueing system is nonempty (Figure 2.9):

$$\boxed{U = 1 - p_0 = \rho \qquad (2.53)}$$

The quantity ρ can also be viewed as the normalized offered load. That is, λ can vary from near zero (light load) to near, but not greater than, μ (heavy load). Thus ρ can vary between 0 and 1.

The state space of the M/M/1 queueing system is infinite in extent. Physically, we are saying that the queueing system has a potentially infinite-sized waiting line. However, since p_n is proportional to ρ^n, the probability of the nth state

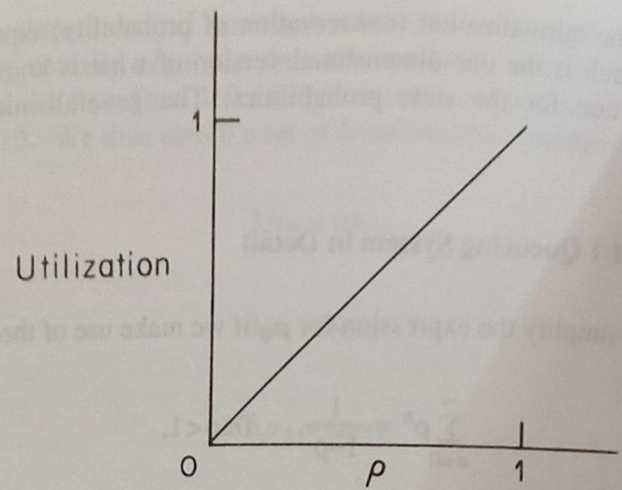

Fig. 2.9: M/M/1 System Utilization

is smaller than the probability of the $n-1$st state by a factor of ρ. Thus the probability of a larger waiting line is smaller than the probability of a smaller waiting line. This can be seen in Figure 2.10 where $\rho = 1/2$.

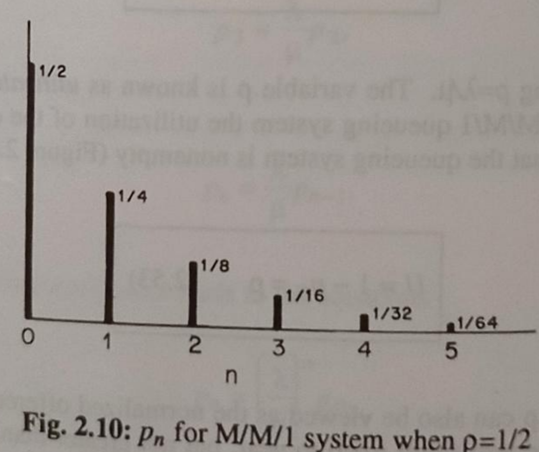

Fig. 2.10: p_n for M/M/1 system when ρ=1/2

Example: Suppose that ρ equals either 0.1, 0.5, or 0.9. Table 2.2 illustrates the probabilities of different waiting line sizes

Table 2.2: Probs. of Queue System Occupancy			
	ρ=.1	ρ=.5	ρ=.9
$P(0\leq\ \leq 3)$	.9999	.9375	.3439
$P(4\leq\ \leq 7)$	1×10^{-4}	.0586	.2256
$P(8\leq\ \leq 11)$	1×10^{-8}	.00366	.1480
$P(\geq 12)$	1×10^{-12}	.000249	.2825

From the table we can see that for a lightly loaded queueing system (ρ=0.1) there are usually less than four customers in the system. When ρ=0.5 the probability that there are less than four customers drops to 94%. At ρ=0.9 it is only 34% and there is a 28% chance of twelve or more customers.

∇

What we have implied in all of this and what we can state directly now is that the arrival rate can never be greater than the service rate for an M/M/1 queueing system. Put another way, ρ can never be greater than one since the queue size would be indefinitely increasing. That is, the queueing system would no longer be in equilibrium.

It is easy to calculate the average number in the queueing system [GROS], [KLEI 75] as follows. The average number is

$$\bar{n} = E(n) = \sum_{n=0}^{\infty} n p_n, \tag{2.54}$$

$$\bar{n} = \sum_{n=0}^{\infty} n(1-\rho)\rho^n, \tag{2.55}$$

$$\bar{n} = (1-\rho)\sum_{n=0}^{\infty} n\rho^n. \tag{2.56}$$

But:

$$\sum_{n=0}^{\infty} n\rho^n = \rho \sum_{n=1}^{\infty} n\rho^{n-1}, \tag{2.57}$$

$$\sum_{n=0}^{\infty} n\rho^n = \rho \frac{d}{d\rho} \sum_{n=0}^{\infty} \rho^n, \tag{2.58}$$

$$\sum_{n=0}^{\infty} n\rho^n = \rho \frac{d}{d\rho} \frac{1}{1-\rho}, \tag{2.59}$$

$$\sum_{n=0}^{\infty} n\rho^n = \frac{\rho}{(1-\rho)^2}. \tag{2.60}$$

So by substituting

$$\boxed{\bar{n} = \frac{\rho}{1-\rho}. \qquad (2.61)}$$

This function is plotted in Figure 2.11. What is most noticeable is the large non-linear increase in the waiting line size as $\rho \to 1$. Interestingly, this function is often used as a "penalty function" in computer network optimization algorithms to penalize large waiting line sizes [GERL].

We can also calculate the variance in this waiting line size. This is defined as

$$\sigma_n^2 = \sum_{n=0}^{\infty} (n-\bar{n})^2 p_n, \tag{2.62}$$

$$\sigma_n^2 = \sum_{n=0}^{\infty} n^2 p_n + \bar{n}^2 \sum_{n=0}^{\infty} p_n - 2\bar{n} \sum_{n=0}^{\infty} n p_n, \tag{2.63}$$

$$\sigma_n^2 = \sum_{n=0}^{\infty} n^2 p_n + \bar{n}^2 - 2\bar{n}^2, \tag{2.64}$$

$$\sigma_n^2 = \sum_{n=0}^{\infty} n^2 p_n - \bar{n}^2. \tag{2.65}$$

But:

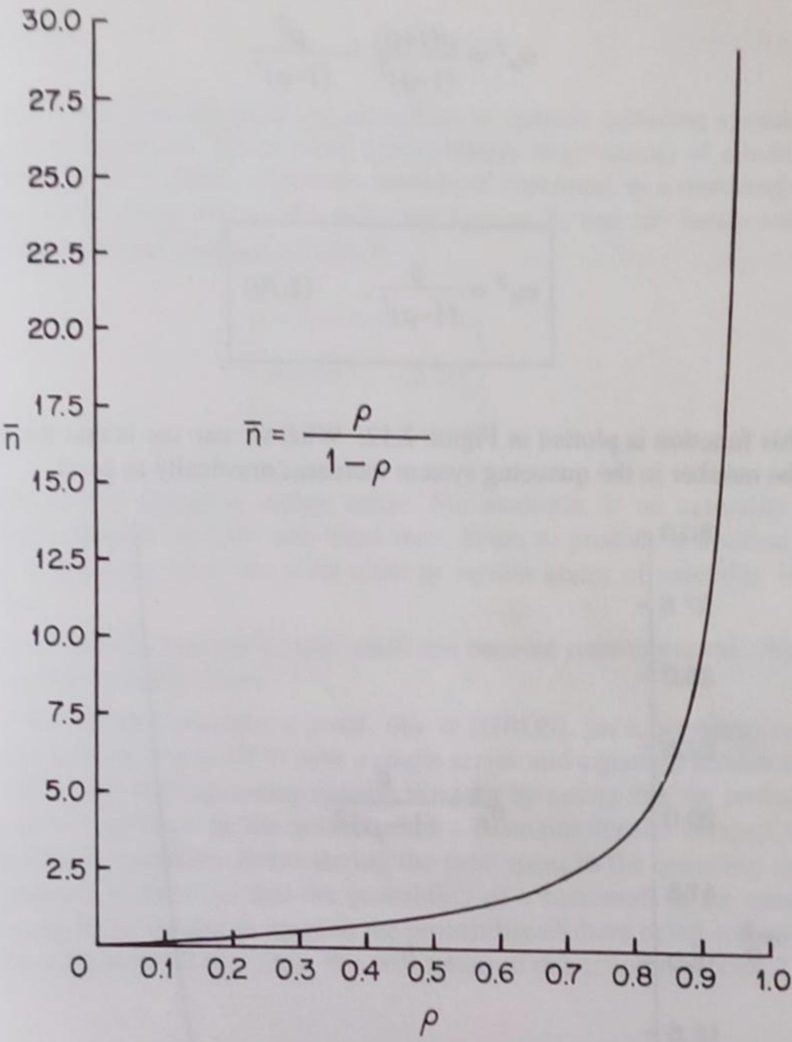

Fig. 2.11: M/M/1: Average Number of Customers

$$\sum_{n=0}^{\infty} n^2 p_n = \sum_{n=0}^{\infty} n^2 (1-\rho)\rho^n, \tag{2.66}$$

$$\sum_{n=0}^{\infty} n^2 p_n = (1-\rho) \sum_{n=0}^{\infty} n^2 \rho^n. \tag{2.67}$$

And:

$$\sum_{n=0}^{\infty} n^2 \rho^n = \frac{\rho(1+\rho)}{(1-\rho)^3}. \tag{2.68}$$

So substituting yields

$$\sigma_n{}^2 = \frac{\rho(1+\rho)}{(1-\rho)^2} - \frac{\rho^2}{(1-\rho)^2}. \tag{2.69}$$

And:

$$\boxed{\sigma_n{}^2 = \frac{\rho}{(1-\rho)^2}. \qquad (2.70)}$$

This function is plotted in Figure 2.12. What we can see is that the variability of the number in the queueing system increases drastically as $\rho \to 1$.

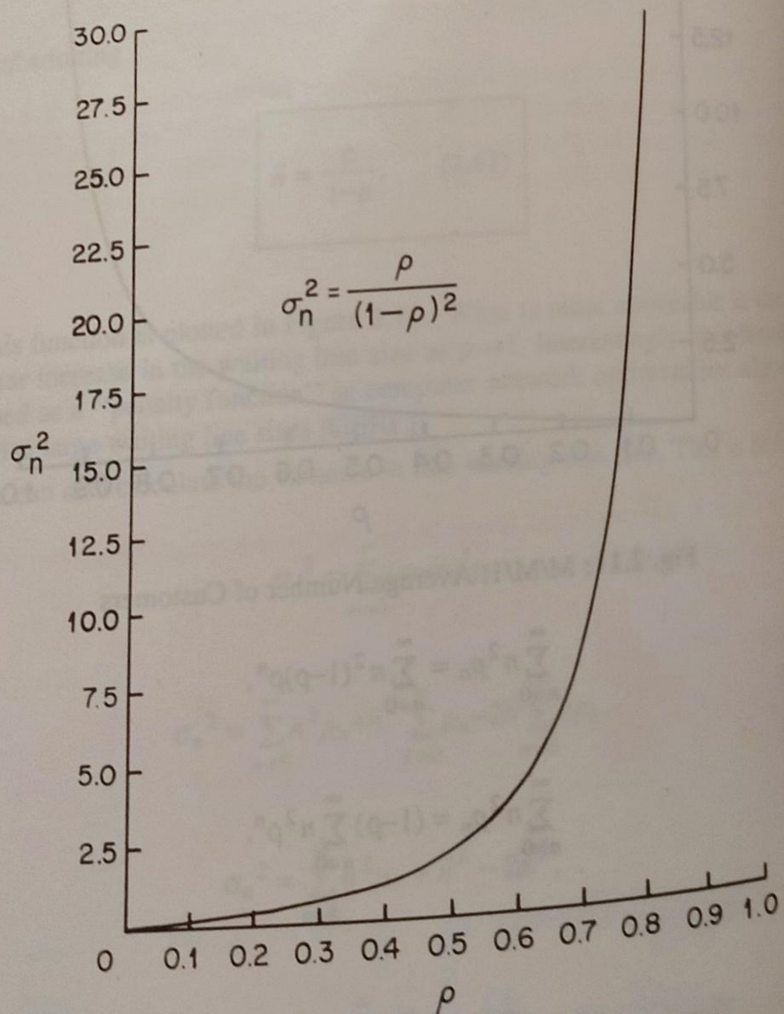

Fig. 2.12: M/M/1: Variance of Number of Customers

3 Little's Law

'e now take some time out from our discussion of specific queueing systems to ok at a simple result that holds under a surprisingly large variety of situations. his is Little's Law. It relates the mean number of customers in a queueing sys- m, $\bar{n}$; the mean arrival rate to the queueing system, λ; and the mean waiting me to get through the queueing system, $\bar{\tau}$:

$$\boxed{\bar{n} = \lambda\bar{\tau}. \qquad (2.71)}$$

Intuitively this equation makes sense. For example, if an assembly line ccepts twenty chassis an hour and takes three hours to produce a finished pro- uct, then at any time there are sixty units in various states of assembly in the ssembly line.

Proofs of this deceptively simple result can become quite involved. We will ive three proofs of Little's Law.

'roof 1: We will first consider a proof, due to [GROS], for a queueing system vith Poisson arrivals that is FIFO with a single server and a general service distri- ution. This is the M/G/1 queueing system. We start by noting that the probability hat there are n customers in the system when a customer departs is equal to the robability that n customers arrive during the time spent in the queueing system y this customer. It turns out that the probability of n customers in the queueing ystem at a departure instant is equal to the probability of there being n customers n the queueing system at any time. We will return to this point in section 2.12.3. So

$$p_n = \int_0^\infty Prob(n \text{ arrivals during } [0,T] \mid T=t)\, d\bar{\tau}(t). \qquad (2.72)$$

Here $\bar{\tau}(t)$ is the cumulative distribution function of waiting time. Using the Pois- son distribution

$$p_n = \int_0^\infty \frac{(\lambda t)^n}{n!} e^{-\lambda t}\, d\bar{\tau}(t). \qquad (2.73)$$

Here $n \geq 0$. Now

$$\bar{n} = E(n) = \sum_{n=0}^{\infty} n p_n. \qquad (2.74)$$

Substituting,

$$\bar{n} = \sum_{n=1}^{\infty} \frac{n}{n!} \int_0^{\infty} (\lambda t)^n e^{-\lambda t} d\bar{\tau}(t). \quad (2.75)$$

The integration and summation can be interchanged:

$$\bar{n} = \int_0^{\infty} d\bar{\tau}(t) \sum_{n=1}^{\infty} e^{-\lambda t} \frac{(\lambda t)^n}{(n-1)!}, \quad (2.76)$$

$$\bar{n} = \int_0^{\infty} \lambda t e^{-\lambda t} d\bar{\tau}(t) \sum_{n=1}^{\infty} \frac{(\lambda t)^{n-1}}{(n-1)!}, \quad (2.77)$$

$$\bar{n} = \int_0^{\infty} \lambda t e^{-\lambda t} d\bar{\tau}(t) \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!}, \quad (2.78)$$

$$\bar{n} = \int_0^{\infty} \lambda t e^{-\lambda t} d\bar{\tau}(t) e^{\lambda t}, \quad (2.79)$$

$$\bar{n} = \lambda \int_0^{\infty} t d\bar{\tau}(t), \quad (2.80)$$

$$\boxed{\bar{n} = \lambda \bar{\tau}. \quad (2.81)}$$

Proof 2: This is a heuristic proof due to P. J. Burke and related by Cooper [COOP]. We start by assuming that the mean values $\bar{n}$ and $\bar{\tau}$ exist and we have an equilibrium situation. We will look at a long interval $[0,t]$. The mean number of customers who arrive is λt. Let each customer bring his/her waiting time with them. The mean amount of waiting time brought in during the interval is $\lambda t \bar{\tau}$.

Now each customer in the queueing system "uses" up waiting time linearly with time. So if $\bar{n}$ is the mean number of customers present during the interval then $\bar{n}t$ is the mean amount of time used up during $[0,t]$. If we let $t \to \infty$, we can expect that the amount of waiting time brought into the queueing system must equal the amount of waiting time used up. Thus

$$\lim_{t\to\infty} \frac{\lambda t\bar{\tau}}{\bar{n}t} = 1 \tag{2.82}$$

and so

$$\boxed{\bar{n} = \lambda\bar{\tau}. \qquad (2.83)}$$

In this heuristic proof no special assumptions were made.

Proof 3: This popular type of proof is based on a diagram like that of Figure 2.13. Versions of it appear in [KOBA], [SCHW 87], [GROS] and [KLEI 75]. We will follow the approach in [KLEI 75]. As in proof 2, we will consider an interval $[0,t]$.

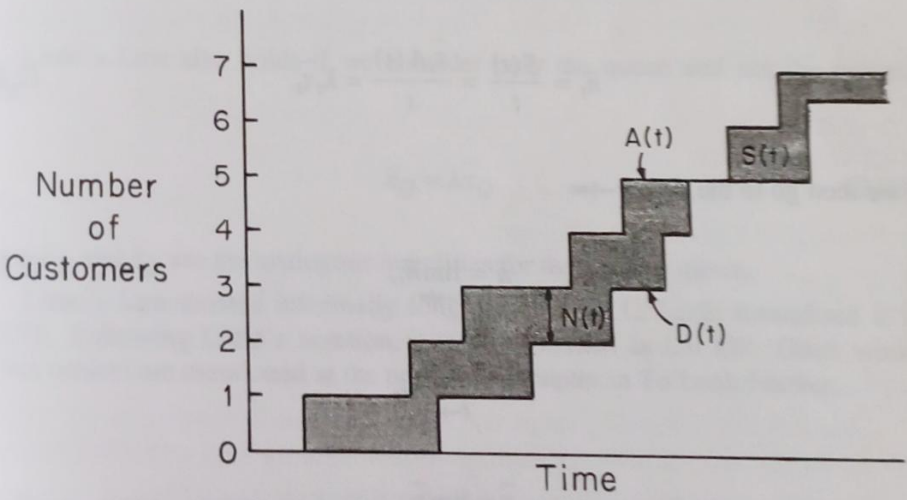

Fig. 2.13: Proving Little's Law

In Figure 2.13 we have $A(t)$ as the number of arrivals in $[0,t]$ and $D(t)$ as the number of departures. Then the number in the system is:

$$N(t) = A(t) - D(t). \tag{2.84}$$

We will now present three simple relations amongst the quantities in Figure 2.13 that can be brought together to form Little's Law. The first is

$$\lambda_t = \frac{A(t)}{t} \tag{2.85}$$

where λ_t is the mean arrival rate during $[0,t]$. The second is

$$\bar{\tau}_t = \frac{S(t)}{A(t)} \qquad (2.86)$$

where $\bar{\tau}_t$ is the mean time spent in the queueing system by each customer and $S(t)$ is the cumulative area between the curves $A(t)$ and $D(t)$ at time t. This area, at t, is the total time spent in the queueing system by all customers up to time t. Finally, the third equation is

$$\bar{n}_t = \frac{S(t)}{t}, \qquad (2.87)$$

which gives the mean number of customers in the queueing system at time t. Now if we substitute these three equations into one another we have

$$\bar{n}_t = \frac{S(t)}{t} = \frac{\bar{\tau}_t A(t)}{t} = \lambda_t \bar{\tau}_t. \qquad (2.88)$$

If we then go to the limit $t \to \infty$

$$\bar{n} = \lim_{t\to\infty} \bar{n}_t, \qquad (2.89)$$

$$\lambda = \lim_{t\to\infty} \lambda_t,$$

$$\bar{\tau} = \lim_{t\to\infty} \bar{\tau}_t.$$

Then

$$\boxed{\bar{n} = \lambda\bar{\tau}. \qquad (2.90)}$$

In this derivation no specific assumptions have been made about the arrival process, the service distribution, the number of servers, or the queueing discipline within the queueing system. Thus Little's Law holds even for a G/G/m queueing system.

Example: Suppose that one wishes to calculate the mean waiting time in an M/M/1 queueing system. We know that

$$\bar{n} = \frac{\rho}{(1-\rho)}, \tag{2.91}$$

so using Little's Law we have

$$\boxed{\bar{\tau} = \frac{1/\mu}{(1-\rho)}. \qquad (2.92)}$$

∇

Little's Law also holds if we consider only the queue and not the server. Then

$$\bar{n}_Q = \lambda\bar{\tau}_Q \tag{2.93}$$

where $\bar{n}_Q$ and $\bar{\tau}_Q$ are the analogous quantities for the waiting queue.

Little's Law existed informally long before J. D. C. Little formalized it in [LITT]. Following Little's notation, it is often written as $L = \lambda W$. Other works on this subject are mentioned at the end of this chapter in To Look Further.

2.4 Reversibility and Burke's Theorem

2.4.1 Introduction

The input to the M/M/1 queueing system is a Poisson process. But what can we say of its output? According to Burke's theorem, which was published in 1956, that output is also Poisson. This result is not immediately intuitive, despite its appealing symmetry. After all, consider the inter-departure times of the M/M/1 queueing system. As long as the queueing system is non-empty, and because we have an exponential server, these are exponentially distributed with mean duration $1/\mu$. But the problem is that sometimes the queueing system *is* empty. Then the time interval between successive departures from the queueing system is the sum of the time it takes for a customer to arrive at the empty queueing system (exponential with mean duration $1/\lambda$) and the time for that customer to be